Politécnico
Castelo Branco
Escola Superior
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Licenciatura em Engenharia Eletrotécnica e das Telecomunicações

Trabalho de Redes de Comunicação de Dados II

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1 INTRODUÇÃO, CONTEXTO E OBJETIVOS

O projeto de RCD2 é constituído por 2 componentes:

- **Componente-A** (15 valores): Trabalho de grupo (2 elementos) relativo à implementação prática, em ambiente de simulação, de uma rede empresarial e que contempla as seguintes tópicos: protocolos de encaminhamento RIPv2 e OSPF (múltipla área), rotas estáticas e agregação de rotas; listas de controlo de acesso (ACLs); protocolos IPv4 e IPv6; VLANs e *trunking*; NAT/PAT estático e dinâmico; protocolo PPP com autenticação PAP e CHAP); tecnologia Frame-Relay; protocolo Spanning-Tree; outros protocolos tais como ARP, ICMP, DHCP, DNS, HTTP, TELNET, SMTP, POP3, FTP, TFTP, *Hot Standby Router Protocol* (HSRP) e EtherChannel (PAgP/LACP); configuração de PCs (clientes e servidores), *switches* e *routers*. A implementação prática tem de ser acompanhada por documentos que suportem o trabalho realizado:
    - 2 Ficheiros de simulação (.pkt);
    - 1 Relatório de acompanhamento (.pdf);
    - 1 Vídeo de acompanhamento (.mp4 ou similar).
- **Componente-B** (5 valores): Prova escrita de avaliação individual.

O projeto simula a implementação de uma rede empresarial composta por 2 zonas geográficas. As zonas estão ligadas por via da Internet e também por via da tecnologia Frame-Relay, como ilustrado na Figura 1.

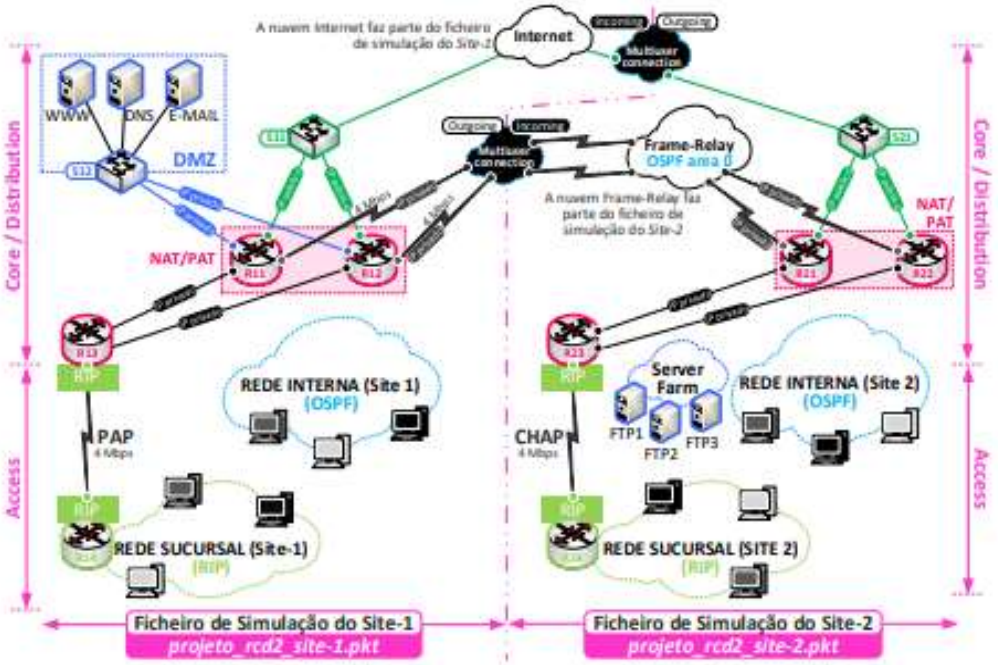
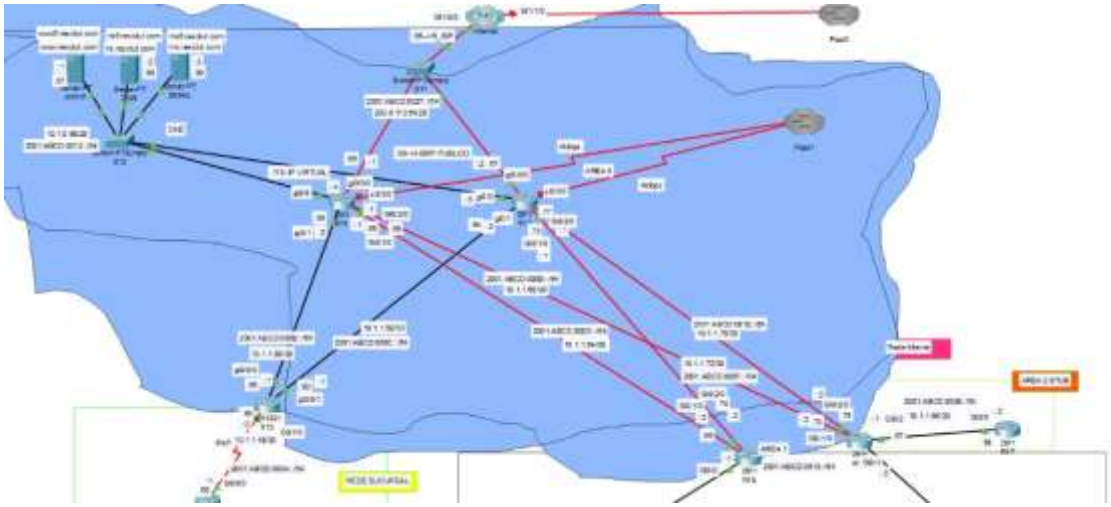


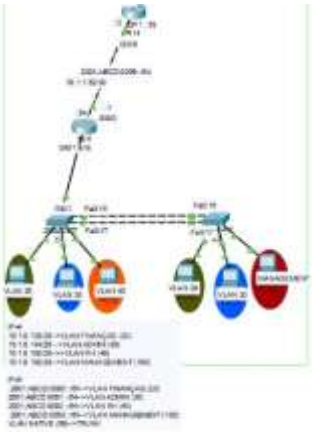
Figura 1 – Topologia simplificada da rede da empresa

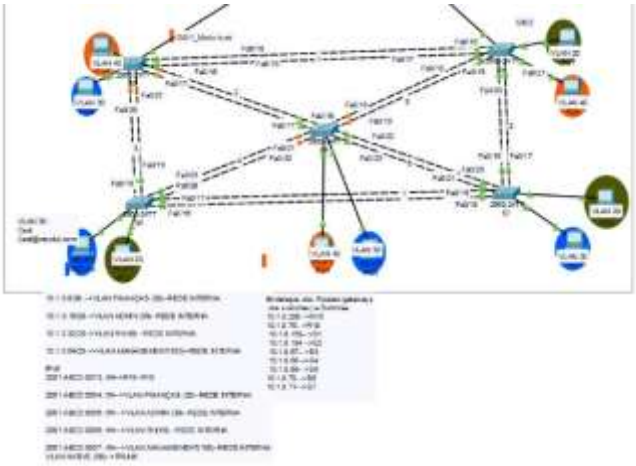
Por análise da Figura 1, observa-se que a implementação em ambiente de simulação irá consistir em dois ficheiros de simulação:

- *projeto_rcd2_CB.pkt*: simula a rede da empresa (*Site-1*) e a Internet.
- *projeto_rcd2_Aveiro.pkt*: simula a rede da empresa (*Site-2*) e a rede Frame-Relay.

Configurações do Site 1







IPv4			
Rede/Sub-rede	Máscara	Localização	Propósito/VLAN
10.1.0.128/28	255.255.255.240	Sucursal Site 1	FINANCAS
10.1.0.144/28	255.255.255.240	Sucursal Site 1	ADMIN
10.1.0.160/28	255.255.255.240	Sucursal Site 1	RH
10.1.0.192/28	255.255.255.240	Sucursal Site 1	MANAGEMENT
10.1.0.195	255.255.255.240	Sucursal Site 1	PCAdministrador
10.1.1.48/30	255.255.255.252	Sucursal Site 1	R13 com R14
10.1.1.52/30	255.255.255.252	Sucursal Site 1	R14 com R15
10.1.1.56/30	255.255.255.252	Rede Inter. Site1	R17 com R16
10.1.1.88/30	255.255.255.252	Site 1	R13 com R11
10.1.1.92/30	255.255.255.252	Site 1	R13 com R12
10.1.1.64/30	255.255.255.252	Site 1	R11 com R18
10.1.1.68/30	255.255.255.252	Site 1	R11 com R16
10.1.1.72/30	255.255.255.252	Site 1	R12 com R18
10.1.1.76/30	255.255.255.252	Site 1	R12 com R16
10.1.0.96/28	255.255.255.240	Site 1	DMZ
10.1.0.0/28	255.255.255.240	Rede Inter.Site1	FINANCAS
10.1.0.16/28	255.255.255.240	Rede Inter.Site1	ADMIN
10.1.0.32/28	255.255.255.240	Rede Inter.Site1	RH
10.1.0.64/28	255.255.255.240	Rede Inter.Site1	MANAGEMENT
203.0.113.64/28	255.255.255.240	Site1	NAT/PAT/Internet

Rede/Sub-rede	Localização	Propósito/VLANs
2001:ABCD:0000::/64	Sucursal Site 1	FINANCAS
2001:ABCD:0001::/64	Sucursal Site 1	ADMIN
2001:ABCD:0002::/64	Sucursal Site 1	RH
2001:ABCD:0003::/64	Sucursal Site 1	MANAGEMENT
2001:ABCD:0004::/64	Rede Interna Site 1	FINANCAS
2001:ABCD:0005::/64	Rede Interna Site 1	ADMIN
2001:ABCD:0006::/64	Rede Interna Site 1	RH
2001:ABCD:0007::/64	Rede Interna Site 1	MANAGEMENT
2001:ABCD:000B::/64	Sucursal Site 1	R13 com R11
2001:ABCD:000C::/64	Sucursal Site 1	R13 com R12
2001:ABCD:000D::/64	Site 1	R11 com R18
2001:ABCD:000E::/64	Site 1	R11 com R16
2001:ABCD:000F::/64	Site 1	R12 com R18
2001:ABCD:0010::/64	Site 1	R12 com R16
2001:ABCD:0011::/64	Site 1	R16 com R17
2001:ABCD:0012::/64	Site 1	DMZ
2001:ABCD:0009::/64	Sucursal Site 1	R14 com R15
2001:ABCD:000A::/64	Sucursal Site 1	R14 com R13
2001:ABCD:0013::/64	Rede Interna Site 1	R18 E R16
2001:ABCD:0027::/64	Site 1	Internet

Abaixo tem a explicação da tabela LSD do router interno da área 0 backbone R11:

1. Temos 5 tipos de LSAs (Link State Datebase);
2. Temos 8 LSAs tipo 1, enviados 1 por cada router na mesma área;
3. Temos 14 LSAs de tipo 2, enviados 1 por cada rede múltiplo acesso com 2 ou routers na mesma área;
4. Temos 12 LSAs de tipo 3, enviados 1 por cada rede externa a esta área, mas no mesmo sistema autónomo (AS);
5. Temos 29 LSAs de tipo 4, enviados 1 por cada router de borda (ABR);
6. Temos 15 LSAs de tipo 5, enviados 1 por cada rede externa a este sistema autónomo;

```
R11#sh ip ospf database
            OSPF Router with ID (11.11.11.11) (Process ID 100)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
22.22.22.22     22.22.22.22     2009        0x80000021 0x004077 10
23.23.23.23     23.23.23.23     819         0x8000000d 0x0048cf 2
21.21.21.21     21.21.21.21     739         0x80000018 0x006b76 10
11.11.11.11     11.11.11.11     93          0x80000014 0x00615b 4
12.12.12.12     12.12.12.12     93          0x80000041 0x0095c8 4
13.13.13.13     13.13.13.13     76          0x8000000d 0x00cc64 2
16.16.16.16     16.16.16.16     75          0x8000000d 0x00e776 2
18.18.18.18     18.18.18.18     67          0x8000000d 0x002e30 2

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.3.137      23.23.23.23     1173        0x80000007 0x00d3a4
10.1.3.157      27.27.27.27     1157        0x80000007 0x00a3d9
10.1.3.145      26.26.26.26     1155        0x80000007 0x002a37
10.1.3.141      23.23.23.23     819         0x80000008 0x002021
10.1.3.165      21.21.21.21     818         0x80000003 0x00f54d
10.1.3.153      27.27.27.27     804         0x80000008 0x00dba0
10.1.3.149      26.26.26.26     804         0x80000008 0x00c55d
10.1.1.89       13.13.13.13     76          0x8000000d 0x001dc0
10.1.1.70       16.16.16.16     75          0x8000000d 0x00def8
10.1.1.66       18.18.18.18     67          0x8000000d 0x00bffd
10.1.1.93       13.13.13.13     65          0x8000000e 0x007e4b
10.1.1.74       18.18.18.18     60          0x8000000e 0x0038cf
10.1.1.78       16.16.16.16     58          0x8000000e 0x0029e7
10.1.0.109      12.12.12.12     57          0x80000007 0x001a5a
```

Figura 1- LSD R11_1º

```
                Summary Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.3.0        26.26.26.26     127         0x800002d8 0x001ff4
10.1.3.16       26.26.26.26     127         0x800002d9 0x007c86
10.1.3.32       26.26.26.26     127         0x800002da 0x00da17
10.1.3.48       26.26.26.26     127         0x800002db 0x0037a9
10.1.3.160      27.27.27.27     124         0x800002e4 0x0090cf
10.1.3.0        27.27.27.27     124         0x800002e5 0x00e61c
10.1.3.16       27.27.27.27     124         0x800002e6 0x0044ad
10.1.3.32       27.27.27.27     124         0x800002e7 0x00a13f
10.1.3.48       27.27.27.27     124         0x800002e8 0x00fed0
10.1.1.56       16.16.16.16     43          0x80000ff4 0x00be1b
10.1.0.0        18.18.18.18     24          0x80000f5a 0x0064b4
10.1.0.0        16.16.16.16     12          0x80001006 0x00e81b

                Summary ASB Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
13.13.13.13     27.27.27.27     1968        0x800002d6 0x00588b
23.23.23.23     27.27.27.27     1967        0x800002d7 0x005669
13.13.13.13     26.26.26.26     1965        0x800002ca 0x008f64
23.23.23.23     26.26.26.26     1964        0x800002cb 0x008c43
12.12.12.12     27.27.27.27     1951        0x800002d8 0x00786e
12.12.12.12     26.26.26.26     1949        0x800002cc 0x00af47
11.11.11.11     11.11.11.11     112         0x80000003 0x0030d3
11.11.11.11     21.21.21.21     1913        0x80000003 0x0003d8
11.11.11.11     23.23.23.23     1913        0x80000005 0x00c20f
11.11.11.11     22.22.22.22     1913        0x80000003 0x00e4f2
21.21.21.21     22.22.22.22     1258        0x80000004 0x001599
21.21.21.21     26.26.26.26     1252        0x800002d0 0x00d4fe
21.21.21.21     27.27.27.27     1252        0x800002dd 0x009c26
22.22.22.22     13.13.13.13     1136        0x80000007 0x00efdb
22.22.22.22     12.12.12.12     1124        0x80000008 0x000cc2
22.22.22.22     23.23.23.23     992         0x80000006 0x00c4df
22.22.22.22     27.27.27.27     987         0x800002e2 0x006455
22.22.22.22     26.26.26.26     987         0x800002d6 0x009a2f
13.13.13.13     16.16.16.16     162         0x80000faa 0x00d45a
13.13.13.13     18.18.18.18     162         0x80000f20 0x00ad04
12.12.12.12     16.16.16.16     142         0x80000fb4 0x00e445
12.12.12.12     18.18.18.18     142         0x80000f27 0x00c3eb
11.11.11.11     12.12.12.12     112         0x80000009 0x0006f3
11.11.11.11     27.27.27.27     112         0x800002e9 0x007a60
11.11.11.11     26.26.26.26     112         0x800002dc 0x00b239
11.11.11.11     18.18.18.18     83          0x80000f3f 0x00c1d9
11.11.11.11     16.16.16.16     83          0x80000fd4 0x00d23b
21.21.21.21     18.18.18.18     71          0x80000f48 0x00afbf
21.21.21.21     16.16.16.16     68          0x80000fe2 0x00b626
```

Figura 2- LSD R11_2º

```
                Type-5 AS External Link States
Link ID         ADV Router      Age         Seq#       Checksum Tag
0.0.0.0         21.21.21.21     1208        0x80000003 0x00a0db 1
0.0.0.0         22.22.22.22     850         0x80000003 0x0082f5 1
10.1.3.96       23.23.23.23     389         0x8000000a 0x0035c6 0
10.1.3.112      23.23.23.23     387         0x8000000a 0x009457 0
10.1.3.132      23.23.23.23     386         0x8000000c 0x0010b9 0
10.1.3.128      23.23.23.23     384         0x8000000c 0x003895 0
10.1.3.80       23.23.23.23     382         0x80000003 0x00e32f 0
10.1.3.64       23.23.23.23     340         0x80000003 0x00849e 0
10.1.1.48       13.13.13.13     109         0x80000007 0x00a8a4 0
10.1.1.52       13.13.13.13     109         0x80000007 0x0080c8 0
10.1.0.128      13.13.13.13     109         0x80000007 0x0048c1 0
10.1.0.144      13.13.13.13     109         0x80000007 0x00a851 0
10.1.0.160      13.13.13.13     109         0x80000007 0x0007e2 0
10.1.0.192      13.13.13.13     109         0x80000007 0x00c504 0
0.0.0.0         12.12.12.12     97          0x80000007 0x00a7f4 1
R11#
```

Figura 3 - LSD R11_3º

```
R11#sh ipv6 ospf database
            OSPF Router with ID (11.11.11.11) (Process ID 10)

                Router Link States (Area 0)

ADV Router      Age         Seq#       Fragment ID  Link count Bits
11.11.11.11     784         0x8000000e 0            4          E
18.18.18.18     786         0x80000008 0            2          B
13.13.13.13     786         0x80000008 0            2          B
16.16.16.16     785         0x80000008 0            2          B
12.12.12.12     783         0x8000000b 0            4

                Net Link States (Area 0)
ADV Router      Age         Seq#       Link ID (DR)   Rtr count
18.18.18.18     791         0x8000000b 6              2
13.13.13.13     791         0x8000000b 1              2
16.16.16.16     790         0x8000000b 7              2
12.12.12.12     788         0x80000006 1              2
18.18.18.18     786         0x8000000c 7              2
13.13.13.13     786         0x8000000c 2              2
16.16.16.16     785         0x8000000c 6              2

                Inter Area Prefix Link States (Area 0)
ADV Router      Age         Seq#       Metric Prefix
13.13.13.13     781         0x8000001f 25     2001:ABCD:A::/64
13.13.13.13     781         0x80000020 26     2001:ABCD:9::/64
16.16.16.16     775         0x8000001f 1      2001:ABCD:13::/64
16.16.16.16     775         0x80000020 1      2001:ABCD:4::/64
16.16.16.16     775         0x80000021 1      2001:ABCD:5::/64
16.16.16.16     775         0x80000022 1      2001:ABCD:6::/64
16.16.16.16     775         0x80000023 1      2001:ABCD:7::/64
18.18.18.18     771         0x8000001a 1      2001:ABCD:13::/64
18.18.18.18     771         0x8000001b 1      2001:ABCD:4::/64
18.18.18.18     771         0x8000001c 1      2001:ABCD:5::/64
18.18.18.18     771         0x8000001d 1      2001:ABCD:6::/64
18.18.18.18     771         0x8000001e 1      2001:ABCD:7::/64
13.13.13.13     771         0x80000021 27     2001:ABCD::/64
13.13.13.13     771         0x80000022 27     2001:ABCD:1::/64
13.13.13.13     771         0x80000023 27     2001:ABCD:2::/64
13.13.13.13     771         0x80000024 27     2001:ABCD:3::/64
16.16.16.16     264         0x80000024 1      2001:ABCD:8::/64
```

Figura 4- LSD IPv6 do R11_ 1º

```
                Link (Type-8) Link States (Area 0)
ADV Router      Age         Seq#       Link ID    Interface
11.11.11.11     828         0x80000019 2          Gi0/1
11.11.11.11     828         0x8000001a 5          Gi0/1/0
11.11.11.11     828         0x8000001b 6          Gi0/2/0
11.11.11.11     799         0x8000001c 1          Gi0/0
13.13.13.13     830         0x8000000d 1          Gi0/1
18.18.18.18     830         0x8000000d 6          Gi0/1/0
16.16.16.16     829         0x8000000d 6          Gi0/2/0
12.12.12.12     798         0x8000001c 1          Gi0/0

                Intra Area Prefix Link States (Area 0)
ADV Router      Age         Seq#       Link ID    Ref-lstype  Ref-LSID
11.11.11.11     828         0x8000000a 2          0x2001      0
18.18.18.18     791         0x8000000b 1          0x2002      7
12.12.12.12     788         0x80000011 1          0x2002      1
12.12.12.12     788         0x80000012 2          0x2001      0
16.16.16.16     783         0x80000011 1          0x2002      6
13.13.13.13     782         0x80000011 1          0x2002      2

                Type-5 AS External Link States
ADV Router      Age         Seq#       Prefix
11.11.11.11     828         0x80000006 ::/0
```

Figura 5- LSD IPv6 do R11_2º

Abaixo tem uma explicação da tabela da LSD (Link State Database) do Router R17, que é interno da área stub:

Cada router OSPF constrói o seu LSD e usa-o para calcular a tabela de rotas.

Uma área stub não permite a propagação de rotas externas, o ABR que interliga esta área bloqueia essa propagação.

1. Para mostrar a LSD usamos o comando show ip ospf database;
2. Temos 2 routers OSPF na área 2, e diz que o router que anunciou e injetou estas redes é o router R16;
3. Temos 3 tipos de LSAs (Link States Advertisements);
4. Temos 2 LSAs de tipo 1, um enviado por ele próprio e o outro pelo R16 (ABR). São os únicos routers que pertencem à área 2 por isso só existem 2 LSAs;
5. Temos 1 LSA de tipo 2, tem 1 por cada rede múltiplo acesso, nesse caso temos dois routers na mesma área;
6. Temos 24 LSAs de tipo 3, que foi injetado pelo ABR (R16) onde temos 1 LSA por cada rede externa a área 2, mas que pertencem o mesmo sistema autónomo (AS);
7. Como a área stub não aprende diretamente todas as rotas externas, o router de borda ABR (R16) injetou uma rota padrão que funciona como um atalho para que se o router interno dessa área envie tráfego para ele (R16) quando precisa ter acesso a redes que não pertencem à área stub.

```
R17#  sh ip ospf database
            OSPF Router with ID (17.17.17.17) (Process ID 100)

                Router Link States (Area 2)

Link ID         ADV Router      Age         Seq#       Checksum Link count
17.17.17.17     17.17.17.17     1519        0x80000008 0x002118 1
16.16.16.16     16.16.16.16     1519        0x80000009 0x0060df 1

                Net Link States (Area 2)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.1.58       17.17.17.17     1519        0x80000006 0x00ffdf

                Summary Net Link States (Area 2)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.1.68       16.16.16.16     1523        0x8000079b 0x00111e
10.1.1.76       16.16.16.16     1503        0x800007ad 0x009c78
0.0.0.0         16.16.16.16     1503        0x800007ae 0x004128
10.1.0.96       16.16.16.16     1494        0x800007b7 0x008c76
10.1.1.92       16.16.16.16     1494        0x800007b8 0x00f008
10.1.1.72       16.16.16.16     1494        0x800007b9 0x00b655
10.1.0.0        16.16.16.16     1458        0x800007c9 0x00d9df
10.1.3.136      16.16.16.16     1155        0x800007d6 0x00b1ff
10.1.3.144      16.16.16.16     1155        0x800007d7 0x005f49
10.1.3.156      16.16.16.16     1155        0x800007d8 0x00e4b6
10.1.3.160      16.16.16.16     1155        0x800007d9 0x007230
10.1.3.140      16.16.16.16     1155        0x800007da 0x008128
10.1.3.152      16.16.16.16     1155        0x800007db 0x000795
10.1.3.148      16.16.16.16     1155        0x800007dc 0x002d72
192.168.2.12    16.16.16.16     1155        0x800007dd 0x005e6c
192.168.2.8     16.16.16.16     1155        0x800007de 0x008449
192.168.2.132   16.16.16.16     1155        0x800007df 0x0002c4
192.168.2.128   16.16.16.16     1155        0x800007e0 0x0028a1
192.168.2.140   16.16.16.16     1155        0x800007e1 0x00ad0f
192.168.2.136   16.16.16.16     1155        0x800007e2 0x00d3eb
10.1.1.88       16.16.16.16     88          0x8000082f 0x003252
10.1.1.64       16.16.16.16     88          0x80000830 0x00217a
192.168.2.0     16.16.16.16     88          0x80000831 0x00f98d
192.168.2.4     16.16.16.16     88          0x80000832 0x00cfb2
R17#  |
```

Figura 6- LSD IPv4 do R17

```
R17#sh ipv6 ospf database
            OSPF Router with ID (17.17.17.17) (Process ID 10)

                Router Link States (Area 2)

ADV Router      Age         Seq#       Fragment ID  Link count Bits
10.1.1.58       447         0x80000001 0            0
17.17.17.17     438         0x80000003 0            1
16.16.16.16     437         0x80000002 0            1          B

                Net Link States (Area 2)
ADV Router      Age         Seq#       Link ID (DR)   Rtr count
17.17.17.17     438         0x80000001 1              2

                Inter Area Prefix Link States (Area 2)
ADV Router      Age         Seq#       Metric Prefix
16.16.16.16     945         0x80000001 1      2001:ABCD:13::/64
16.16.16.16     945         0x80000002 1      2001:ABCD:4::/64
16.16.16.16     945         0x80000003 1      2001:ABCD:5::/64
16.16.16.16     945         0x80000004 1      2001:ABCD:6::/64
16.16.16.16     945         0x80000005 1      2001:ABCD:7::/64
16.16.16.16     945         0x80000006 1      ::/0
16.16.16.16     945         0x80000007 2      2001:ABCD:B::/64
16.16.16.16     945         0x80000008 2      2001:ABCD:D::/64
16.16.16.16     945         0x80000009 2      2001:ABCD:12::/64
16.16.16.16     945         0x8000000a 3      2001:ABCD:F::/64
16.16.16.16     945         0x8000000b 2      2001:ABCD:10::/64
16.16.16.16     945         0x8000000c 2      2001:ABCD:C::/64
16.16.16.16     945         0x8000000d 1      2001:ABCD:E::/64
16.16.16.16     940         0x8000000e 27     2001:ABCD:A::/64
16.16.16.16     940         0x8000000f 28     2001:ABCD:9::/64
16.16.16.16     940         0x80000010 29     2001:ABCD::/64
16.16.16.16     940         0x80000011 29     2001:ABCD:1::/64
16.16.16.16     940         0x80000012 29     2001:ABCD:2::/64
16.16.16.16     940         0x80000013 29     2001:ABCD:3::/64

                Link (Type-8) Link States (Area 2)
ADV Router      Age         Seq#       Link ID    Interface
10.1.1.58       447         0x80000001 1          Gi0/0
17.17.17.17     442         0x80000002 1          Gi0/0
16.16.16.16     438         0x80000002 3          Gi0/0

                Intra Area Prefix Link States (Area 2)
ADV Router      Age         Seq#       Link ID    Ref-lstype  Ref-LSID
10.1.1.58       447         0x80000001 2          0x2001      0
17.17.17.17     442         0x80000002 1          0x2002      1
16.16.16.16     960         0x80000003 1          0x2002      3
R17#
```

Figura 7- LSD IPv6 do R17

Aí abaixo se seguem as configurações finais de todos os Routers do Site 1:

```
hostname R15
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ip dhcp pool FINANCAS
network 10.1.0.128 255.255.255.240
default-router 10.1.0.142
dns-server 10.1.0.98
domain-name financas.resolut.com
ip dhcp pool ADMIN
network 10.1.0.144 255.255.255.240
default-router 10.1.0.158
dns-server 10.1.0.98
domain-name admin.resolut.com
ip dhcp pool RH
network 10.1.0.160 255.255.255.240
default-router 10.1.0.174
dns-server 10.1.0.98
domain-name rh.resolut.com
!
ipv6 unicast-routing
!
ipv6 dhcp pool VLAN-FINANCAS
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
ipv6 dhcp pool VLAN-ADMIN
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
ipv6 dhcp pool VLAN-RH
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
```

```
ipv6 dhcp pool VLAN-MANAGEMENT
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
!
interface GigabitEthernet0/0
description #LIGA A REDE DA SUCURSAL #
ip address 10.1.1.54 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::15 link-local
ipv6 address 2001:ABCD:9::1/64
ipv6 ospf 10 area 3
!
interface GigabitEthernet0/1
description #LIGA A REDE SUCURSAL - VLANS#
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/1.20
encapsulation dot1Q 20
ip address 10.1.0.142 255.255.255.240
ipv6 address 2001:ABCD::/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 3
ipv6 dhcp server VLAN-FINANCAS
!
interface GigabitEthernet0/1.30
encapsulation dot1Q 30
ip address 10.1.0.158 255.255.255.240
ipv6 address 2001:ABCD:1::/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 3
ipv6 dhcp server VLAN-RH
```

```
!
interface GigabitEthernet0/1.40
encapsulation dot1Q 40
ip address 10.1.0.174 255.255.255.240
ipv6 address 2001:ABCD:2::/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 3
ipv6 dhcp server VLAN-RH
!
interface GigabitEthernet0/1.99
encapsulation dot1Q 99 native
no ip address
!
interface GigabitEthernet0/1.100
encapsulation dot1Q 100
ip address 10.1.0.206 255.255.255.240
ipv6 address 2001:ABCD:3::/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 3
ipv6 dhcp server VLAN-MANAGEMENT
!
router rip
version 2
network 10.0.0.0
no auto-summary
!
ipv6 router ospf 10
router-id 15.15.15.15
log-adjacency-changes
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
access-class 10 in
```

```
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
end

!
hostname R14
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ipv6 unicast-routing
!
username R13 password 0 redes2
!
interface GigabitEthernet0/0
description #LIGA A REDE DA SUCURSAL #
ip address 10.1.1.53 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::14 link-local
ipv6 address 2001:ABCD:9::2/64
ipv6 ospf 10 area 3
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
description #LIGACAO COM PAP #
```

```
bandwidth 4000
ip address 10.1.1.50 255.255.255.252
encapsulation ppp
ppp authentication pap
ppp pap sent-username R14 password 0 parents
ipv6 address FE80::14 link-local
ipv6 address 2001:ABCD:A::1/64
ipv6 ospf 10 area 3
router rip
version 2
network 10.0.0.0
no auto-summary
!
ipv6 router ospf 10
router-id 14.14.14.14
log-adjacency-changes
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
End
```

```
!
hostname R13
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ipv6 unicast-routing
!
no ipv6 cef
!
username R14 password 0 parents
!
interface GigabitEthernet0/0/0
description #LIGA A AREA 0 #
ip address 10.1.1.89 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::13 link-local
ipv6 address 2001:ABCD:B::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/0/1
description #LIGACAO PAP#
ip address 10.1.1.93 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::13 link-local
ipv6 address 2001:ABCD:C::1/64
ipv6 ospf 10 area 0
!
interface Serial0/1/0
bandwidth 4000
ip address 10.1.1.49 255.255.255.252
encapsulation ppp
ppp authentication pap
ppp pap sent-username R13 password 0 redes2
```

```
ipv6 address FE80::13 link-local
ipv6 address 2001:ABCD:A::2/64
ipv6 ospf 10 area 3
clock rate 4000000
!
router ospf 100
router-id 13.13.13.13
log-adjacency-changes
redistribute rip subnets
auto-cost reference-bandwidth 1000
network 10.1.1.88 0.0.0.7 area 0
!
router rip
version 2
redistribute ospf 100 metric 12
network 10.0.0.0
no auto-summary
!
ipv6 router ospf 10
router-id 13.13.13.13
log-adjacency-changes
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
login
!
end
```

```
!
hostname R11
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ipv6 unicast-routing
!
license udi pid CISCO2911/K9 sn FTX1524VK76-
!
interface GigabitEthernet0/0
description #LIGA A DMZ#
ip address 10.1.0.108 255.255.255.240
ip nat inside
duplex auto
speed auto
ipv6 address FE80::11 link-local
ipv6 address 2001:ABCD:12::4/64
ipv6 ospf 10 area 0
standby 1 ip 10.1.0.110
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/1
description #LIGA A AREA 0 - ROUTER Q INTERLIGA A REDE SUCURSAL#
ip address 10.1.1.90 255.255.255.252
ip nat inside
duplex auto
speed auto
ipv6 address FE80::11 link-local
ipv6 address 2001:ABCD:B::2/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/0/0
description #LIGA A INTERNET#
ip address 203.0.113.66 255.255.255.240
```

```
ip nat outside
ipv6 address FE80::11 link-local
ipv6 address 2001:ABCD:27::1/64
standby 2 ip 203.0.113.68
standby 2 priority 110
standby 2 preempt
!
interface GigabitEthernet0/1/0
description #LIGA A REDE INTERNA - AREA 0#
ip address 10.1.1.65 255.255.255.252
ip nat inside
ipv6 address FE80::11 link-local
ipv6 address 2001:ABCD:D::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2/0
description #LIGA A REDE INTERNA - AREA 0 #
ip address 10.1.1.69 255.255.255.252
ip nat inside
ipv6 address FE80::11 link-local
ipv6 address 2001:ABCD:E::1/64
ipv6 ospf 10 area 0
!
interface Serial0/3/0
description #LIGACAO COM O SITE 2 #
no ip address
encapsulation frame-relay ietf
frame-relay lmi-type q933a
!
interface Serial0/3/0.101 point-to-point
description #LIGA A R11 com o DCLI R11 para R21#
bandwidth 4000
ip address 192.168.2.5 255.255.255.252
frame-relay interface-dlci 101
clock rate 2000000
```

```
!
interface Serial0/3/0.102 point-to-point
description #LIGA A R11 com o DCLI R11 para R22#
bandwidth 4000
ip address 192.168.2.1 255.255.255.252
frame-relay interface-dlci 102
clock rate 2000000
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 100
router-id 11.11.11.11
log-adjacency-changes
auto-cost reference-bandwidth 1000
network 10.1.1.64 0.0.0.31 area 0
network 10.1.0.108 0.0.0.0 area 0
network 192.168.2.0 0.0.0.255 area 0
default-information originate
!
ipv6 router ospf 10
router-id 11.11.11.11
default-information originate
log-adjacency-changes
!
ip nat pool SITE1 203.0.113.72 203.0.113.78 netmask 255.255.255.240
ip nat inside source list 1 pool SITE1 overload
ip nat inside source static tcp 10.1.0.97 80 203.0.113.69 80
ip nat inside source static tcp 10.1.0.98 53 203.0.113.70 53
```

```
ip nat inside source static udp 10.1.0.98 53 203.0.113.70 53
ip nat inside source static tcp 10.1.0.99 25 203.0.113.71 25
ip nat inside source static tcp 10.1.0.99 110 203.0.113.71 110
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.65
!
ipv6 route ::/0 2001:ABCD:27::3
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
access-list 1 remark #ACL PARA TRADUCAO NAT-PAT#
access-list 1 permit 10.1.0.0 0.0.1.255
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
end
```

!

hostname R12

```
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ipv6 unicast-routing
!
interface GigabitEthernet0/0
description #LIGA A DMZ#
ip address 10.1.0.109 255.255.255.240
ip nat inside
```

```
duplex auto
speed auto
ipv6 address FE80::12 link-local
ipv6 address 2001:ABCD:12::5/64
ipv6 ospf 10 area 0
standby 1 ip 10.1.0.110
!
interface GigabitEthernet0/1
description #LIGA A AREA 0 - ROUTER Q INTERLIGA A REDE SUCURSAL#
ip address 10.1.1.94 255.255.255.252
ip nat inside
duplex auto
speed auto
ipv6 address FE80::12 link-local
ipv6 address 2001:ABCD:C::2/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/0/0
description #LIGA A INTERNET #
ip address 203.0.113.67 255.255.255.240
ip nat outside
ipv6 address FE80::11 link-local
ipv6 address 2001:ABCD:27::2/64
standby 2 ip 203.0.113.68
!
interface GigabitEthernet0/1/0
description #LIGA A REDE INTERNA - AREA 0#
ip address 10.1.1.73 255.255.255.252
ip nat inside
ipv6 address FE80::12 link-local
ipv6 address 2001:ABCD:F::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2/0
description #LIGA A REDE INTERNA - AREA 0#
```

```
ip address 10.1.1.77 255.255.255.252
ip nat inside
ipv6 address FE80::12 link-local
ipv6 address 2001:ABCD:10::1/64
ipv6 ospf 10 area 0
!
interface Serial0/3/0
description #LIGACAO COM O SITE 2 #
bandwidth 4000
no ip address
encapsulation frame-relay ietf
frame-relay lmi-type q933a
!
interface Serial0/3/0.103 point-to-point
description #LIGA A R12 com o DCLI R22 para R11#
bandwidth 4000
ip address 192.168.2.13 255.255.255.252
frame-relay interface-dlci 103
clock rate 2000000
!
interface Serial0/3/0.104 point-to-point
description #LIGA A R12 com o DCLI R12 para R21#
bandwidth 4000
ip address 192.168.2.9 255.255.255.252
frame-relay interface-dlci 104
clock rate 2000000
!
router ospf 100
router-id 12.12.12.12
log-adjacency-changes
auto-cost reference-bandwidth 1000
network 10.1.0.109 0.0.0.0 area 0
network 10.1.1.64 0.0.0.31 area 0
network 192.168.2.0 0.0.0.255 area 0
default-information originate
```

```
!
ipv6 router ospf 10
router-id 12.12.12.12
log-adjacency-changes
!
ip nat pool SITE1 203.0.113.72 203.0.113.78 netmask 255.255.255.240
ip nat inside source list 1 pool SITE1 overload
ip nat inside source static tcp 10.1.0.97 80 203.0.113.69 80
ip nat inside source static tcp 10.1.0.98 53 203.0.113.70 53
ip nat inside source static udp 10.1.0.98 53 203.0.113.70 53
ip nat inside source static tcp 10.1.0.99 25 203.0.113.71 25
ip nat inside source static tcp 10.1.0.99 110 203.0.113.71 110
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.65
!
ip flow-export version 9
!
ipv6 route ::/0 2001:ABCD:27::3
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
access-list 1 remark #ACL PARA TRADUCAO NAT-PAT#
access-list 1 permit 10.1.0.0 0.0.1.255
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
end
!
```

```
!
hostname R18
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ip dhcp pool FINANCAS
network 10.1.0.0 255.255.255.240
default-router 10.1.0.14
dns-server 10.1.0.98
domain-name financas.resolut.com
ip dhcp pool ADMIN
network 10.1.0.16 255.255.255.240
default-router 10.1.0.30
dns-server 10.1.0.98
domain-name admin.resolut.com
ip dhcp pool RH
network 10.1.0.32 255.255.255.240
default-router 10.1.0.46
dns-server 10.1.0.98
domain-name rh.resolut.com
!
ipv6 unicast-routing
!
ipv6 dhcp pool VLAN-FINANCAS
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
ipv6 dhcp pool VLAN-ADMIN
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
ipv6 dhcp pool VLAN-RH
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
```

```
interface GigabitEthernet0/0
description #LIGA A REDE INTERNA - AREA 1 - VLANS#
no ip address
duplex auto
speed auto
ipv6 address 2001:ABCD:13::1/64
ipv6 ospf 10 area 1
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 10.1.0.13 255.255.255.240
ipv6 address 2001:ABCD:4::5/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 1
ipv6 dhcp server VLAN-FINANCAS
standby 1 ip 10.1.0.14
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 10.1.0.29 255.255.255.240
ipv6 address 2001:ABCD:5::9/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 1
ipv6 dhcp server VLAN-ADMIN
standby 1 ip 10.1.0.30
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 10.1.0.45 255.255.255.240
ipv6 address 2001:ABCD:6::D/64
ipv6 nd other-config-flag
```

```
ipv6 ospf 10 area 1
ipv6 dhcp server VLAN-RH
standby 1 ip 10.1.0.46
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.99
encapsulation dot1Q 99 native
no ip address
standby 1 preempt
!
interface GigabitEthernet0/0.100
encapsulation dot1Q 100
ip address 10.1.0.76 255.255.255.240
ipv6 address 2001:ABCD:7::11/64
ipv6 ospf 10 area 1
standby 1 ip 10.1.0.77
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/1/0
description #LIGA A REDE INTERNA - AREA 0 #
ip address 10.1.1.66 255.255.255.252
ipv6 address FE80::18 link-local
ipv6 address 2001:ABCD:D::2/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2/0
description #LIGA A REDE INTERNA - AREA 0 #
ip address 10.1.1.74 255.255.255.252
ipv6 address FE80::18 link-local
ipv6 address 2001:ABCD:F::2/64
ipv6 ospf 10 area 0
!
router ospf 100
```

```
router-id 18.18.18.18
log-adjacency-changes
area 1 range 10.1.0.0 255.255.255.128
auto-cost reference-bandwidth 1000
network 10.1.0.0 0.0.0.15 area 1
network 10.1.0.16 0.0.0.15 area 1
network 10.1.0.32 0.0.0.15 area 1
network 10.1.0.64 0.0.0.15 area 1
network 10.1.1.64 0.0.0.15 area 0
!
ipv6 router ospf 10
router-id 18.18.18.18
log-adjacency-changes
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
End
```

```
!
hostname R16
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
ip dhcp pool FINANCAS
network 10.1.0.0 255.255.255.240
default-router 10.1.0.14
dns-server 10.1.0.98
domain-name financas.resolut.com
ip dhcp pool ADMIN
network 10.1.0.16 255.255.255.240
default-router 10.1.0.30
dns-server 10.1.0.98
domain-name admin.resolut.com
ip dhcp pool RH
network 10.1.0.32 255.255.255.240
default-router 10.1.0.46
dns-server 10.1.0.98
domain-name rh.resolut.com
!
ipv6 unicast-routing
!
ipv6 dhcp pool VLAN-FINANCAS
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
ipv6 dhcp pool VLAN-ADMIN
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
ipv6 dhcp pool VLAN-RH
dns-server 2001:ABCD:12::2
domain-name resolut.com
!
```

```
interface GigabitEthernet0/1
description #LIGA A REDE INTERNA - AREA 1 - VLANS#
no ip address
duplex auto
speed auto
ipv6 address 2001:ABCD:13::2/64
ipv6 ospf 10 area 1
!
interface GigabitEthernet0/1.20
encapsulation dot1Q 20
ip address 10.1.0.12 255.255.255.240
ipv6 address 2001:ABCD:4::6/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 1
ipv6 dhcp server VLAN-FINANCAS
standby 1 ip 10.1.0.14
!
interface GigabitEthernet0/1.30
encapsulation dot1Q 30
ip address 10.1.0.28 255.255.255.240
ipv6 address 2001:ABCD:5::A/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 1
ipv6 dhcp server VLAN-ADMIN
standby 1 ip 10.1.0.30
!
interface GigabitEthernet0/1.40
encapsulation dot1Q 40
ip address 10.1.0.44 255.255.255.240
ipv6 address 2001:ABCD:6::E/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 1
ipv6 dhcp server VLAN-RH
standby 1 ip 10.1.0.46
!
```

```
interface GigabitEthernet0/1.99
encapsulation dot1Q 99 native
no ip address
!
interface GigabitEthernet0/1.100
encapsulation dot1Q 100
ip address 10.1.0.75 255.255.255.240
ipv6 address 2001:ABCD:7::12/64
ipv6 nd other-config-flag
ipv6 ospf 10 area 1
standby 1 ip 10.1.0.77
!
interface GigabitEthernet0/2
description #LIGA A REDE INTERNA - AREA STUB #
ip address 10.1.1.57 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::16 link-local
ipv6 address 2001:ABCD:8::1/64
ipv6 ospf 10 area 2
!
interface GigabitEthernet0/1/0
description #LIGA A REDE INTERNA - AREA 0#
ip address 10.1.1.70 255.255.255.252
ipv6 address FE80::16 link-local
ipv6 address 2001:ABCD:E::2/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2/0
description #LIGA A REDE INTERNA - AREA 0#
ip address 10.1.1.78 255.255.255.252
ipv6 address FE80::16 link-local
ipv6 address 2001:ABCD:10::2/64
ipv6 ospf 10 area 0
!
```

```
router ospf 100
router-id 16.16.16.16
log-adjacency-changes
area 1 range 10.1.0.0 255.255.255.128
area 2 stub
auto-cost reference-bandwidth 1000
network 10.1.0.0 0.0.0.15 area 1
network 10.1.0.16 0.0.0.15 area 1
network 10.1.0.32 0.0.0.15 area 1
network 10.1.0.64 0.0.0.15 area 1
network 10.1.1.57 0.0.0.0 area 2
network 10.1.1.64 0.0.0.15 area 0
!
ipv6 router ospf 10
router-id 16.16.16.16
log-adjacency-changes
area 2 stub
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
end
```

```
!
hostname R17
!
ipv6 unicast-routing
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface GigabitEthernet0/0
description #LIGA A REDE INTERNA - AREA STUB #
ip address 10.1.1.58 255.255.255.252
duplex auto
speed auto
ipv6 address 2001:ABCD:0008::2/64
 ipv6 address FE80::17 link-local
 ipv6 ospf 10 area 2
!
router ospf 100
router-id 17.17.17.17
log-adjacency-changes
area 2 stub
auto-cost reference-bandwidth 1000
network 10.1.1.58 0.0.0.0 area 2
!
ipv6 router ospf 10
 area 2 stub
 router-id 17.17.17.17
log-adjacency-changes
area 2 stub
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
access-class 10 in
password Site1
login
line vty 5 15
access-class 10 in
password Site1
login
!
end
```

```
!
hostname R_ISP
!
ipv6 unicast-routing
!
interface FastEthernet0/0
description #LIGA AO GOOGLE#
ip address 209.85.135.254 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2A00:1450:4004:801::1/64
!
interface Ethernet0/0/0
description #LIGA AO POTE DE MEL#
ip address 198.0.0.254 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2A03:2880:2130:CF05::1/64
!
interface Ethernet0/1/0
description *LIGA A TLD DNS .COM*
ip address 192.5.6.254 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2001:503:A83E::2:1/64
!
interface Ethernet0/2/0
description *LIGA A TLD DNS .ORG*
ip address 199.19.56.254 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
```

```
ipv6 address 2001:500:E::2/64
!
interface FastEthernet1/0
description *LIGA A ROOT DNS*
ip address 193.0.14.254 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2001:420:1201:6::1/64
!
interface FastEthernet1/1
description *LIGA A PC NA INTERNET*
ip address 223.1.1.254 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2001:500:88:200::1/64
!
interface GigabitEthernet1/0/0
description #LIGACAO COM O SITE 1 #
ip address 203.0.113.65 255.255.255.240
ip access-group 100 in
ipv6 address FE80::1 link-local
ipv6 address 2001:ABCD:27::3/64
!
interface GigabitEthernet1/1/0
description #LIGA O SITE 2#
ip address 203.0.113.87 255.255.255.240
ip access-group 100 in
ipv6 address 2001:ABCD:26::20/64
!
interface Vlan1
no ip address
shutdown
!
```

```
ip classless
ip route 22.2.2.0 255.255.255.0 22.2.2.3
ip route 0.0.0.0 0.0.0.0 203.0.113.68
ip route 0.0.0.0 0.0.0.0 203.0.113.83
!
ipv6 route ::/0 2001:ABCD:27::1
ipv6 route ::/0 2001:ABCD:26::21
!
access-list 100 remark *ACL QUE PROIBE ENDERECOS PRIVADOS RFC 1918*
access-list 100 deny ip 10.0.0.0 0.255.255.255 any
access-list 100 deny ip 172.16.0.0 0.15.255.255 any
access-list 100 deny ip 192.168.0.0 0.0.255.255 any
access-list 100 permit ip any any
!
banner motd ^CO ALUNO DEVE CONFIGURAR AS DUAS INTERFACES (gi1/0/0 e gi1/1/0)
    QUE LIGAM AO SITE-1 E SITE-2. O ALUNO DEVE TAMBEM CONFIGURAR
    ENCAMINHAMENTO ESTATICO POR FORMA A QUE ESTE ROUTER CONSIGA
    ENCAMINHAR PACOTES PARA OS DOIS SITES DA EMPRESA^C
end
```

Aí abaixo se seguem as configurações finais de todos os switches do Site1 que possuem VLANs:

hostname S2

```
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
```

```
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
```

```
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode auto
!
interface FastEthernet0/17
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode auto
!
interface FastEthernet0/18
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/19
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/20
 switchport trunk native vlan 99
```

```
 switchport mode trunk
!
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.0.194 255.255.255.240
!
ip default-gateway 10.1.0.206
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 password Site1
 login
end
```

```
!
hostname S1
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
```

```
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface FastEthernet0/17
 switchport trunk native vlan 99
```

```
 switchport mode trunk
 channel-group 1 mode desirable
!
interface FastEthernet0/18
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/19
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/20
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
```

```
interface Vlan100
 ip address 10.1.0.193 255.255.255.240
!
ip default-gateway 10.1.0.206
!

access-list 10 permit host 10.1.0.195
access-list 10 deny any
line con 0
!
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 password Site1
 login

!
End
```

```
!
hostname S3
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel2
 switchport trunk native vlan 99
 switchport mode trunk
```

```
!
interface Port-channel3
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
```

```
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface FastEthernet0/17
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface FastEthernet0/18
```

```
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface FastEthernet0/19
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface FastEthernet0/20
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
```

```
interface Vlan100
 ip address 10.1.0.67 255.255.255.240
!
ip default-gateway 10.1.0.76
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 login
!
End
```

```
!
hostname S4
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel2
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel3
 switchport trunk native vlan 99
 switchport mode trunk
```

```
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
```

```
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode auto
!
interface FastEthernet0/17
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode auto
!
interface FastEthernet0/18
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
```

```
interface FastEthernet0/19
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface FastEthernet0/20
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/2
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.0.68 255.255.255.240
!
ip default-gateway 10.1.0.76
```

```
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 login
!
end
```

---

hostname S5

```
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel2
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel3
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel4
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel5
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
```

```
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
```

```
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode auto
!
interface FastEthernet0/17
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode auto
!
interface FastEthernet0/18
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode auto
!
interface FastEthernet0/19
 switchport trunk native vlan 99
```

```
 switchport mode trunk
 channel-group 3 mode auto
!
interface FastEthernet0/20
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode desirable
!
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode desirable
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode desirable
!
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode desirable
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.0.69 255.255.255.240
!
ip default-gateway 10.1.0.76
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
```

```
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 login
!
end
```

---

hostname S6

```
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel3
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel4
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
```

```
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
```

```
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface FastEthernet0/17
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface FastEthernet0/18
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode auto
!
interface FastEthernet0/19
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode auto
!
interface FastEthernet0/20
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode auto
!
```

```
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode auto
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.0.70 255.255.255.240
!
ip default-gateway 10.1.0.76
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 password Site1
 login
!
end
```

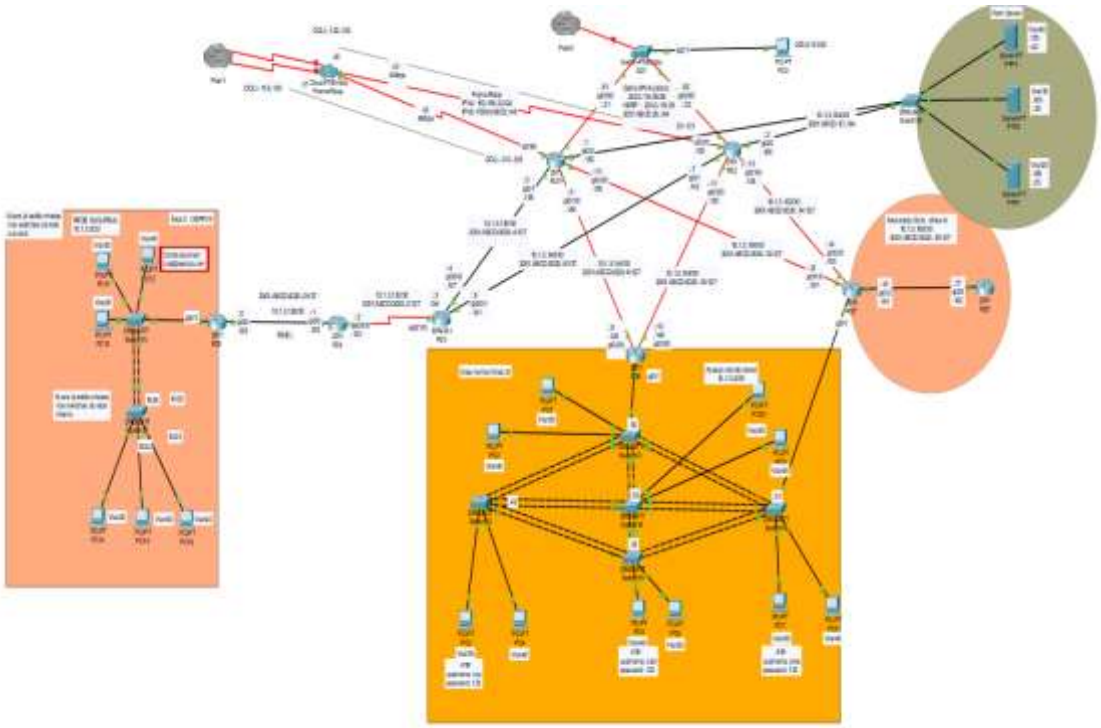
```
hostname S7
!
enable secret 5 $1$mERr$DLJAmWivmLetw5mTkuQ6u/
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel2
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel5
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 100
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 100
 switchport mode access
!
```

```
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 40
 switchport mode access
```

```
!
interface FastEthernet0/16
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode auto
!
interface FastEthernet0/17
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode auto
!
interface FastEthernet0/18
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode auto
!
interface FastEthernet0/19
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode auto
!
interface FastEthernet0/20
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode auto
!
interface FastEthernet0/21
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode auto
!
interface FastEthernet0/22
 switchport trunk native vlan 99
 switchport mode trunk
!
```

```
interface FastEthernet0/23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.0.71 255.255.255.240
!
ip default-gateway 10.1.0.76
!
access-list 10 permit host 10.1.0.195
access-list 10 deny any
!
line vty 0 4
 access-class 10 in
 password Site1
 login
line vty 5 15
 access-class 10 in
 login
```

Configurações do site 2



Rede interna – SITE2			
VLAN	**Sub-rede**	**IPs utilizáveis**	**Broadcast**
VLAN 20 - FINANCAS	10.1.3.0/28	10.1.3.1 – 10.1.3.14	10.1.3.15
VLAN 30 - ADMIN	10.1.3.16/28	10.1.3.17-10.1.23.30	10.1.3.31
VLAN 40 – RH	10.1.3.32/28	10.1.3.33 – 10.1.3.46	10.1.3.47
VLAN 100 -MANAGMENT	10.1.3.48/28	10.1.3.49 – 10.1.3.62	10.1.3.63
Rede da sucursal– SITE2			
VLAN 20 - FINANCAS	10.1.3.0/28	10.1.3.1 – 10.1.3.14	10.1.3.15
VLAN 30 - ADMIN	10.1.3.16/28	10.1.3.17 – 10.1.3.30	10.1.3.31
VLAN 40 - RH	10.1.3.32/28	10.1.3.33 – 10.1.3.46	10.1.3.47
VLAN 100 -MANAGMENT	10.1.3.48/28	10.1.3.49 – 10.1.3.62	10.1.3.63

	RA-RB	Sub-rede	Roteador A	Roteador B
1	R25-24	10.4.3.128/30	10.4.3.129	10.4.3.130
2	R24-23	10.4.3.132/30	10.4.3.133	10.4.3.134

	RA-RB	Sub-rede	Roteador A	Roteador B
3	R23-21	10.4.3.136/30	10.4.3.137	10.4.3.138
4	R23-22	10.4.3.140/30	10.4.3.141	10.4.3.142
5	R26-R21	10.4.3.144/30	10.4.3.145	10.4.3.146
6	R26-R22	10.4.3.148/30	10.4.3.149	10.4.3.150
7	R22-27	10.4.3.152/30	10.4.3.153	10.4.3.154
8	R27-R21	10.4.3.156/30	10.4.3.157	10.4.3.158
9	R27-R28	10.4.3.160/30	10.4.3.161	10.4.3.162
Server Farm				
		Sub-rede	**IP**	
1	R21-SW19	10.4.3.164/28	10.4.3.165	
2	R22- SW19	10.4.3.164/28	10.4.3.166	
Endereços dos servidores IPV4				
	Servidores	**Sub-rede**	**IP dos Servers**	
1	FTP1	10.4.3.164/28	10.4.3.168	
2	FTP2	10.4.3.164/28	10.4.3.169	
3	FTP3	10.4.3.164/28	10.4.3.170	
NAT				
	IP das interfaces	Endereços usados para tradução		
R21	203.0.113.81	203.0.113.81 - 203.0.113.81/28		
R22	203.0.113.82	203.0.113.82- 203.0.113.81/28		
Frame-Relay				
	Links Seriais	**RA**	**RB**	
1	R11-R21	192.168.2.5/30	192.168.2.6/30	
2	R11-R22	192.168.2.1/30	192.168.2.2/30	
3	R12-R21	192.168.2.9/30	192.168.2.10/30	
4	R12-R22	192.168.2.13/30	192.168.2.14/30	

Link entre os Router IPV6			
Links	**Prefixo /127**	**Roteador A**	**Roteador B**
R25-R24	2001:ABCD:0020::0/127	2001:ABCD:0020::0	2001:ABCD:0020::1
R24-R23	2001:ABCD:0020::2/127	2001:ABCD:0020::2	2001:ABCD:0020::3
R23-R21	2001:ABCD:0020::4/127	2001:ABCD:0020::4	2001:ABCD:0020::5
R23-R22	2001:ABCD:0020::6/127	2001:ABCD:0020::6	2001:ABCD:0020::7
R26-R21	2001:ABCD:0020::8/127	2001:ABCD:0020::8	2001:ABCD:0020::9
R26-R22	2001:ABCD:0020::10/127	2001:ABCD:0020::10	2001:ABCD:0020::11
R27-R21	2001:ABCD:0020::12/127	2001:ABCD:0020::12	2001:ABCD:0020::13
R27-R22	2001:ABCD:0020::14/127	2001:ABCD:0020::14	2001:ABCD:0020::15
R27-R28	2001:ABCD:0020::16/127	2001:ABCD:0020::17	2001:ABCD:0020::18

Tabela do LSD do Router interna da da área Totally Stub

```
R28>en
R28#sh ip ospf database
            OSPF Router with ID (28.28.28.28) (Process ID 100)

                Router Link States (Area 4)

Link ID         ADV Router      Age         Seq#       Checksum Link count
28.28.28.28     28.28.28.28     1561        0x80000005 0x00d539 1
27.27.27.27     27.27.27.27     1564        0x80000006 0x001501 1

                Net Link States (Area 4)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.3.162      28.28.28.28     1561        0x80000003 0x000ee0

                Summary Net Link States (Area 4)
Link ID         ADV Router      Age         Seq#       Checksum
0.0.0.0         27.27.27.27     1559        0x80000003 0x00628d
R28# S
```

Figura 8- Tabela LSD R28_Totally stub

O ABR (R27) transmite apenas uma rota por defeito que permite o R28 chegar a redes que fazem parte de outras áreas ou AS's. O R28 tem a tabela de encaminhamento bastante pequena pelo que permitem apenas o LSA do tipo 1 que é gerado por ele mesmo e LSA do tipo 3 que é gerado pelo R27.

```
!
hostname S12
enable password cisco
!
interface range FastEthernet0/4 - 0/6
 switchport access vlan 100
 switchport mode access
!
interface range FastEthernet0/7 - 0/9
 switchport access vlan 20
 switchport mode access
!
interface range FastEthernet0/10 - 0/12
 switchport access vlan 30
 switchport mode access
!
interface range FastEthernet0/13 - 0/15
 switchport access vlan 40
 switchport mode access
!
```

```
interface range FastEthernet0/16 - 0/24
switchport trunk native vlan 99
GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.3.49 255.255.255.240
!
ip default-gateway 10.1.3.62
line con 0
line vty 0 4password cisco
login
line vty 5 15
login
!
hostname S13
enable password cisco
!
interface range FastEthernet0/4 - 0/6
 switchport access vlan 100
 switchport mode access
!
interface range FastEthernet0/7 - 0/9
 switchport access vlan 20
 switchport mode access
!
interface range FastEthernet0/10 - 0/12
 switchport access vlan 30
 switchport mode access
!
interface range FastEthernet0/13 - 0/15
 switchport access vlan 40
```

```
 switchport mode access
!
interface range FastEthernet0/16 - 0/24, GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan100
 ip address 10.1.3.50 255.255.255.240
ip default-gateway 10.1.3.62
line con 0
line vty 0 4
 password cisco
 login
line vty 5 15
 login

!
hostname S14
enable password cisco
!
interface range FastEthernet0/4 - 0/6
 switchport access vlan 100
 switchport mode access
!
interface range FastEthernet0/7 - 0/9
 switchport access vlan 20
 switchport mode access
!
interface range FastEthernet0/10 - 0/12
 switchport access vlan 30
 switchport mode access
```

```
!
interface range FastEthernet0/13 - 0/15
 switchport access vlan 40
 switchport mode access
!
interface range FastEthernet0/16 - 0/24, GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
!
ip address 10.1.3.51 255.255.255.240
ip default-gateway 10.1.3.62
line con 0
line vty 0 4
 password cisco
 login
line vty 5 15
 login

!
hostname S15
enable password cisco
!
interface range FastEthernet0/4 - 0/6
 switchport access vlan 100
 switchport mode access
!
interface range FastEthernet0/7 - 0/9
 switchport access vlan 20
 switchport mode access
!
interface range FastEthernet0/10 - 0/12
 switchport access vlan 30
```

```
 switchport mode access
!
interface range FastEthernet0/13 - 0/15
 switchport access vlan 40
 switchport mode access
!
interface range FastEthernet0/16 - 0/24
switchport trunk native vlan 99
GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.3.49 255.255.255.240
!
ip default-gateway 10.1.3.62
line con 0
line vty 0 4password cisco
login
line vty 5 15
login
!
!
hostname S18
enable password cisco
!
interface range FastEthernet0/4 - 0/6
 switchport access vlan 100
 switchport mode access
!
interface range FastEthernet0/7 - 0/9
 switchport access vlan 20
 switchport mode access
```

```
!
interface range FastEthernet0/10 - 0/12
 switchport access vlan 30
 switchport mode access
!
interface range FastEthernet0/13 - 0/15
 switchport access vlan 40
 switchport mode access
!
interface range FastEthernet0/16 - 0/24
switchport trunk native vlan 99
GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan100
 ip address 10.1.3.49 255.255.255.240
!
ip default-gateway 10.1.3.62
line con 0
line vty 0 4password cisco
login
line vty 5 15
login
!
```

```
!
hostname R25
ip dhcp pool FINANCAS
network 10.1.3.64 255.255.255.240
default-router 10.1.3.78
dns-server 10.1.0.98
domain-name financas.resolut.com
ip dhcp pool ADMIN
network 10.1.3.80 255.255.255.240
default-router 10.1.3.94
dns-server 10.1.0.98
domain-name admin.resolut.com
ip dhcp pool RH
network 10.1.3.96 255.255.255.240
default-router 10.1.3.110
dns-server 10.1.0.98
domain-name rh.resolut.com
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
```

```
!
interface GigabitEthernet0/0
description #LIGA AO ROUTER R24#
ip address 10.1.3.129 255.255.255.252
ipv6 address FE80::25 link-local
```

```
ipv6 address 2001:ABCD:20::/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1
description #LIGA A REDE SUCURSAL#
!
interface GigabitEthernet0/1.20
encapsulation dot1Q 20
ip address 10.1.3.78 255.255.255.240
ipv6 address 2001:ABCD:18::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1.30
encapsulation dot1Q 30
ip address 10.1.3.94 255.255.255.240
ipv6 address 2001:ABCD:19::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1.40
encapsulation dot1Q 40
ip address 10.1.3.110 255.255.255.240
ipv6 address 2001:ABCD:1A::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1.99
encapsulation dot1Q 99 native
no ip address
!
interface GigabitEthernet0/1.100
encapsulation dot1Q 100
ip address 10.1.3.113 255.255.255.240
ipv6 address 2001:ABCD:1B::1/64
ipv6 ospf 10 area 0
```

```
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 10.0.0.0
no auto-summary
!
ipv6 router ospf 10
router-id 25.25.25.25
line con 0
!
line aux 0
!
line vty 0 4
login
!
End
```

```
!
hostname R24
!
no ip cef
ipv6 unicast-routing
!
username R23 password 0 cisco000
interface Loopback0
ip address 24.24.24.24 255.255.255.255
!
interface GigabitEthernet0/0
description #LIGA AO ROUTER DA SUCURSAL R25#
ip address 10.1.3.130 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::26 link-local
ipv6 address 2001:ABCD:20::1/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/3/0
description #LIGA AO R23#
```

```
bandwidth 4000
ip address 10.1.3.133 255.255.255.252
encapsulation ppp
ppp authentication chap
ipv6 address FE80::24 link-local
ipv6 address 2001:ABCD:20::2/127
ipv6 ospf 10 area 0
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 100
log-adjacency-changes
!
router rip
version 2
network 10.0.0.0
no auto-summary
!
ipv6 router ospf 10
line con 0
!
line aux 0
!
line vty 0 4
login
end
```

```
!
hostname R23
!
ipv6 unicast-routing
!
username R24 password 0 cisco000
!
interface Loopback0
ip address 23.23.23.23 255.255.255.255
!
interface GigabitEthernet0/0/0
description #LIGA AO ROUTER ATIVO R21 AREA 0#
ip address 10.1.3.137 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::24 link-local
ipv6 address 2001:ABCD:20::4/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/0/1
description #LIGA AO ROUTER STANDBY R22 AREA 0#
ip address 10.1.3.141 255.255.255.252
duplex auto
speed auto
ipv6 address FE80::23 link-local
ipv6 address 2001:ABCD:20::6/127
ipv6 ospf 10 area 0
!
interface Serial0/1/0
ip address 10.1.3.134 255.255.255.252
encapsulation ppp
ppp authentication chap
ipv6 address FE80::23 link-local
```

```
ipv6 address 2001:ABCD:20::3/127
ipv6 ospf 10 area 0
clock rate 2000000
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown

router ospf 100
router-id 23.23.23.23
redistribute rip subnets
auto-cost reference-bandwidth 1000
network 10.1.3.136 0.0.0.7 area 0
!
router rip
version 2
redistribute ospf 100 metric 2
network 10.0.0.0
no auto-summary
!
ipv6 router ospf 10
router-id 23.23.23.23
line con 0
!
line aux 0
!
line vty 0 4
login!
End
```

```
!
hostname R21
ipv6 unicast-routing
!
no ipv6 cef
!
interface GigabitEthernet0/0
description #LIGA AO SWITCH SW19 SERVER FARM#
ip address 10.1.3.165 255.255.255.240
duplex auto
speed auto
ipv6 address 2001:ABCD:1C::1/64
ipv6 ospf 10 area 0
standby version 2
standby 1 ip 10.1.3.167
standby 1 priority 150
!
interface GigabitEthernet0/1
description #LIGA AO ROUTER R23#
ip address 10.1.3.138 255.255.255.252
ip nat inside
duplex auto
speed auto
ipv6 address FE80::23 link-local
ipv6 address 2001:ABCD:20::5/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
```

```
interface GigabitEthernet0/0/0
description #LIGACAO PARA A INTERNET#
ip address 203.0.113.81 255.255.255.240
ip nat outside
ipv6 address 2001:ABCD:26::21/64
standby version 2
standby 20 ip 203.0.113.83
standby 20 priority 150
standby 20 preempt
!
interface GigabitEthernet0/1/0
description #LIGA AO ROUTER R26 AREA 2 REDE INTERNA#
ip address 10.1.3.146 255.255.255.252
ip nat inside
ipv6 address FE80::21 link-local
ipv6 address 2001:ABCD:20::9/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2/0
description #LIGA AO ROUTER R27 #
ip address 10.1.3.158 255.255.255.252
ip nat inside
ipv6 address FE80::21 link-local
ipv6 address 2001:ABCD:20::13/127
ipv6 ospf 10 area 0
!
interface Serial0/3/0
description #LIGACAO PARA O FRAME-RELAY#
bandwidth 4000
no ip address
encapsulation frame-relay ietf
frame-relay lmi-type q933a
ip nat outside
```

```
!
interface Serial0/3/0.112 point-to-point
description #LIGA A R21 com o DCLI R11 para R21#
ip address 192.168.2.133 255.255.255.252
frame-relay interface-dlci 112
clock rate 2000000
!
interface Serial0/3/0.201 point-to-point
description #LIGA A R21 com o DCLI R21 para R11#
bandwidth 4000
ip address 192.168.2.6 255.255.255.252
frame-relay interface-dlci 201
clock rate 2000000
!
interface Serial0/3/0.204 point-to-point
description #LIGA A R21 com o DCLI R21 para R12#
bandwidth 4000
ip address 192.168.2.10 255.255.255.252
frame-relay interface-dlci 204
clock rate 2000000
!
interface Serial0/3/0.211 point-to-point
description #LIGA A R21 com o DCLI R21 para R11#
ip address 192.168.2.129 255.255.255.252
frame-relay interface-dlci 211
clock rate 2000000
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
```

```
router ospf 100
router-id 21.21.21.21
log-adjacency-changes
auto-cost reference-bandwidth 1000
network 10.1.3.128 0.0.0.31 area 0
network 10.1.3.164 0.0.0.3 area 0
network 192.168.2.0 0.0.0.255 area 0
default-information originate
!
ipv6 router ospf 10
router-id 21.21.21.21
log-adjacency-changes
redistribute connected
!
ipv6 router ospf 1
log-adjacency-changes
redistribute connected
!
ip nat pool REDEINTERNA 203.0.113.81 203.0.113.95 netmask 255.255.255.240
ip nat inside source list 1 pool REDEINTERNA overload
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.87
!
ip flow-export version 9
!
ipv6 route ::/0 2001:ABCD:26::20
access-list 1 permit 10.1.2.0 0.0.1.255
line con 0
!
line vty 0 4
login
!
End
```

```
!
hostname R22
!
ipv6 unicast-routing
!
interface GigabitEthernet0/0
description #LIGA AO SWITCH SW19 SERVER FARM#
ip address 10.1.3.166 255.255.255.240
duplex auto
speed auto
ipv6 address 2001:ABCD:1C::2/64
ipv6 ospf 10 area 0
standby version 2
standby 1 ip 10.1.3.167
!
interface GigabitEthernet0/1
description #LIGA AO ROUTER R23 REDE SUCURSAL#
ip address 10.1.3.142 255.255.255.252
ip nat inside
duplex auto
speed auto
ipv6 address FE80::22 link-local
ipv6 address 2001:ABCD:20::7/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/0/0
description #LIGACAO PARA A INTERNET#
```

```
ip address 203.0.113.82 255.255.255.240
ip nat outside
ipv6 address 2001:ABCD:26::22/64
standby version 2
standby 20 ip 203.0.113.83
!
interface GigabitEthernet0/1/0
description #LIGA AO ROUTER R27#
ip address 10.1.3.154 255.255.255.252
ip nat inside
ipv6 address FE80::22 link-local
ipv6 address 2001:ABCD:20::15/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2/0
description #LIGA AO ROUTER R26 AREA 2 REDE INTERNA#
ip address 10.1.3.150 255.255.255.252
ip nat inside
ipv6 address FE80::22 link-local
ipv6 address 2001:ABCD:20::11/127
ipv6 ospf 10 area 0
!
interface Serial0/3/0
description #LIGACAO FRAME-RELAY#
bandwidth 4000
no ip address
encapsulation frame-relay ietf
frame-relay lmi-type q933a
ip nat outside
!
interface Serial0/3/0.122 point-to-point
description #LIGA A R22 com o DCLI R12 para R22#
ip address 192.168.2.141 255.255.255.252
```

```
frame-relay interface-dlci 122
clock rate 2000000
!
interface Serial0/3/0.202 point-to-point
description #LIGA A R22 com o DCLI R22 para R11#
bandwidth 4000
ip address 192.168.2.2 255.255.255.252
frame-relay interface-dlci 202
clock rate 2000000
!
interface Serial0/3/0.203 point-to-point
description #LIGA A R22 com o DCLI R22 para R12#
bandwidth 4000
ip address 192.168.2.14 255.255.255.252
frame-relay interface-dlci 203
clock rate 2000000
!
interface Serial0/3/0.221 point-to-point
description #LIGA A R22 com o DCLI R22 para R12#
ip address 192.168.2.137 255.255.255.252
frame-relay interface-dlci 221
clock rate 2000000
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 100
```

```
router-id 22.22.22.22
log-adjacency-changes
auto-cost reference-bandwidth 1000
network 10.1.3.164 0.0.0.3 area 0
network 10.1.3.128 0.0.0.31 area 0
network 192.168.2.0 0.0.0.255 area 0
default-information originate
!
router rip
!
ipv6 router ospf 10
router-id 22.22.22.22
log-adjacency-changes
redistribute connected
!
ipv6 router ospf 1
redistribute connected
!
ip nat pool REDEINTERNA 203.0.113.82 203.0.113.95 netmask 255.255.255.240
ip nat inside source list 1 pool REDEINTERNA overload
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.87
!
access-list 1 permit 10.1.2.0 0.0.1.255
!
line con 0
!
line vty 0 4
login
end
```

```
!
hostname R26
!
ip dhcp excluded-address 10.1.3.13 10.1.3.14
ip dhcp excluded-address 10.1.3.29 10.1.3.30
ip dhcp excluded-address 10.1.3.45 10.1.3.46
ip dhcp excluded-address 10.1.3.61 10.1.3.62
!
ip dhcp pool FINANCAS
network 10.1.3.0 255.255.255.240
default-router 10.1.3.14
dns-server 10.1.0.98
domain-name financas.resolut.com
ip dhcp pool ADMIN
network 10.1.3.16 255.255.255.240
default-router 10.1.3.30
dns-server 10.1.0.98
domain-name admin.resolut.com
ip dhcp pool RH
network 10.1.3.32 255.255.255.240
default-router 10.1.3.46
dns-server 10.1.0.98
domain-name rh.resolut.com
!
no ip cef
ipv6 unicast-routing
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
shutdown
!
```

```
interface GigabitEthernet0/1
description #LIGA AO SWITCH S13 DA AREA 2#

!
interface GigabitEthernet0/1.20
encapsulation dot1Q 20
ip address 10.1.3.13 255.255.255.240
ipv6 address 2001:ABCD:14::1/64
ipv6 ospf 10 area 2
standby version 2
standby 1 ip 10.1.3.14
standby 1 priority 150
standby 1 preempt
!
interface GigabitEthernet0/1.30
encapsulation dot1Q 30
ip address 10.1.3.29 255.255.255.240
ipv6 address 2001:ABCD:15::1/64
ipv6 ospf 10 area 2
standby version 2
standby 1 ip 10.1.3.30
standby 1 priority 150
standby 1 preempt
!
interface GigabitEthernet0/1.40
encapsulation dot1Q 40
ip address 10.1.3.45 255.255.255.240
ipv6 address 2001:ABCD:16::1/64
ipv6 ospf 10 area 2
standby version 2
standby 1 ip 10.1.3.46
standby 1 priority 150
standby 1 preempt
```

```
!
interface GigabitEthernet0/1.99
encapsulation dot1Q 99 native
no ip address
!
interface GigabitEthernet0/1.100
encapsulation dot1Q 100
ip address 10.1.3.61 255.255.255.240
ipv6 address 2001:ABCD:17::1/64
ipv6 ospf 10 area 2
standby version 2
standby 1 ip 10.1.3.62
standby 1 priority 150
standby 1 preempt
!
interface GigabitEthernet0/2/0
description #LIGA AO ROUTER ATIVO R21#
ip address 10.1.3.145 255.255.255.252
ipv6 address FE80::26 link-local
ipv6 address 2001:ABCD:20::8/127
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/3/0
description #LIGA AO ROUTER STANDBY R22#
ip address 10.1.3.149 255.255.255.252
ipv6 address FE80::26 link-local
ipv6 address 2001:ABCD:20::10/127
ipv6 ospf 10 area 0
!
router ospf 100
router-id 26.26.26.26
auto-cost reference-bandwidth 1000
network 10.1.3.144 0.0.0.7 area 0
```

```
network 10.1.3.0 0.0.0.63 area 2
!
ipv6 router ospf 10
router-id 26.26.26.26
line con 0
!
line vty 0 4
login
!
End
```

```
!
hostname R28
!
interface Loopback0
ip address 21.21.21.21 255.255.255.255
!
interface GigabitEthernet0/0
description #LIGA AO ROUTER R27#
ip address 10.1.3.162 255.255.255.252
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
```

```
!
interface Vlan1
no ip address
shutdown
!
router ospf 100
router-id 28.28.28.28
log-adjacency-changes
area 4 stub no-summary
auto-cost reference-bandwidth 1000
network 10.1.3.162 0.0.0.0 area 4
!
```